



Abstract

This work focuses on the design, fabrication, and experimental validation of a low-voltage electromagnetic generator intended for a small-scale wind turbine system. All calculations were performed using SI units to ensure consistency and traceability of results in accordance with international engineering standards [1].

A key contribution was the end-to-end development of the generator system, including CAD modelling in Onshape [2], iterative design refinement, and cross-team integration via Microsoft Teams [3]. Initial conceptual collaboration evolved into full individual ownership of the generator design, including coil configuration, magnetic layout considerations, and mechanical assembly.

Preliminary circuit modelling was developed using LTspice [4] to support subsystem integration and guide early electrical design decisions. Electromagnetic coil design was informed by Faraday's Law of Electromagnetic Induction [5], combined with experimentally validated adjustments to refine turn count and winding configuration.

Prototype fabrication and assembly were conducted in a workshop environment, followed by controlled testing using a drill-driven rotor and multimeter instrumentation, later extended through Arduino Uno [6] data acquisition. Experimental outcomes were systematically compared against theoretical predictions, enabling identification of key performance limitations including air-gap losses, resistive heating, and back EMF effects.

Overall, the study demonstrates the iterative nature of electromechanical design, where theoretical models required significant correction when confronted with real-world electromagnetic and mechanical constraints.



Introduction

The primary technical responsibility undertaken in this project was the complete design, construction, and validation of the electromagnetic generator subsystem. Early-stage design discussions led to the proposal of a two-stage spur gear system intended to increase rotor angular velocity and therefore improve induced EMF output. This decision was grounded in the synchronous speed relationship:

$$N_s = \frac{60f}{p}$$

where f represents electrical frequency and p denotes pole pairs of the stator configuration.

The rationale behind the gear ratio selection was to overcome the limitation of low rotational input speeds from the wind turbine and ensure operation within a voltage range sufficient to achieve the target 6V output. However, subsequent testing revealed that mechanical stability constraints and electromagnetic losses were not fully captured in the initial analytical assumptions.

Although the CAD model provided a structurally clear assembly pathway, practical implementation highlighted significant deviations between idealised and real conditions. In particular, mechanical instability of the gear system and vibration-induced misalignment introduced inefficiencies not predicted in the theoretical design stage.

The final assembled system produced only partial voltage output, primarily limited by air-gap flux leakage, mechanical vibration, and unmodelled resistive and back EMF effects.



Methods

The methodology followed an iterative engineering design approach combining basic theoretical modelling, prototype development, and experimental testing to develop a small-scale electromagnetic generator.

The process began with a review of relevant literature on small generator design, focusing on coil windings, magnet arrangements, and typical sources of inefficiency such as resistive losses and air-gap effects. This informed the selection of a practical design concept suitable for fabrication and testing within workshop constraints.

A baseline generator configuration was then developed using CAD software (Onshape) and supporting calculations based on Faraday's Law of Electromagnetic Induction. Magnetic field estimates were used alongside these calculations to determine an initial range for the number of coil turns required to achieve an output close to 6V under ideal conditions. Simplifying assumptions were applied at this stage, including uniform magnetic flux and negligible electrical and mechanical losses. A prototype was subsequently manufactured in the workshop using the CAD model as a reference. The generator consisted of a rotor with permanent magnets and a stator containing copper wire coils arranged according to the initial design. Care was taken during assembly to ensure alignment between rotor and stator, although tolerances were limited by manual fabrication methods.

Initial testing was carried out by mechanically driving the rotor using a handheld drill to provide controlled rotation. Output voltage was measured across the coil terminals using a multimeter. These tests were used to compare actual performance against theoretical predictions. Following initial results, performance deviations were identified. These were linked to Back EMF, coil resistance, air-gap variation, and mechanical issues such as vibration and rotor imbalance. Based on this, the design was revised by reducing the stator from eight coils to four and producing six different coil winding configurations with varying numbers of turns for comparison.

Further testing used the same drill-driven setup, with rotational speed gradually increased for each configuration. Voltage readings were recorded consistently under each condition to allow comparison between designs and to identify the most effective coil arrangement.

The results were then compared to identify which configuration most closely achieved the target voltage under practical operating conditions. The method ensured that each design decision was validated through physical testing rather than relying solely on theoretical assumptions.

Results, Analysis and Discussion

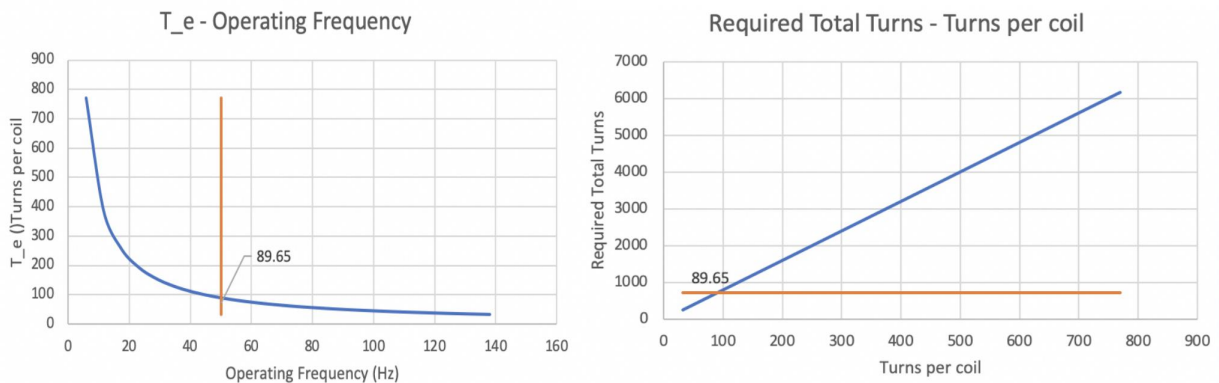


Figure 1: Theoretical coil design model.

Left: relationship between operating frequency and induced electrical output under idealised assumptions.

Right: calculated requirement for total number of coil turns and corresponding turns per coil configuration.

Initial modelling assumed an idealised electromagnetic system consisting of 8 uniformly distributed rectangular magnets. Under these conditions, coil winding calculations suggested a feasible voltage output within the target range. However, this model did not incorporate several critical non-ideal factors, including:

- Back electromotive force (Back EMF) opposing induced current
- Resistive heating losses in the copper windings
- Flux leakage across the air gap
- Mechanical vibration-induced misalignment

As a result, the theoretical model overestimated practical voltage output capability, demonstrating a significant deviation between ideal electromagnetic theory and physical implementation constraints.

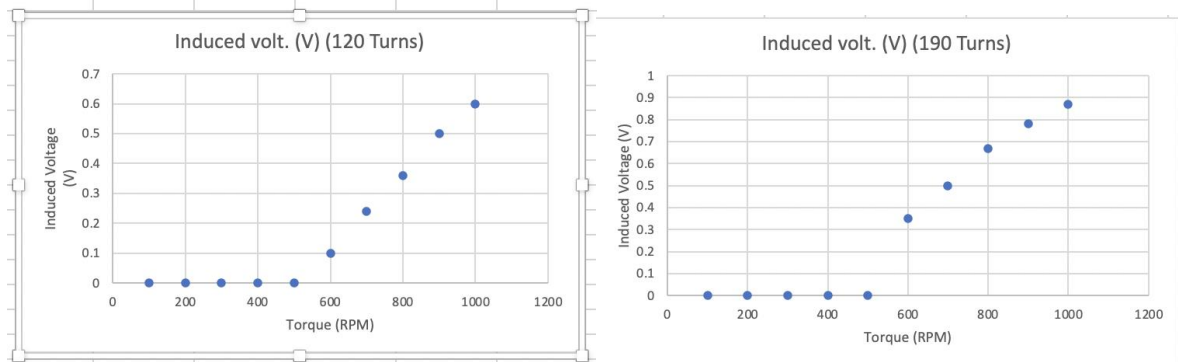


Figure 2 Comparison of the experimental voltage output data.

Experimental results indicate that a configuration of approximately 190 turns per coil in a 4-coil arrangement produced the highest observed voltage output, approaching but not fully achieving the 6V design target under optimal rotational speed conditions.

Key observations include:

- e) Voltage generation exhibited a non-linear relationship with rotational speed, consistent with Faraday's Law.
- f) A threshold rotational speed was required before measurable output was observed, attributed to the need to overcome internal resistance and Back EMF.
- g) Coil resistance significantly limited current flow, reducing effective power transfer despite increased induced EMF at higher RPM.

The experimental data demonstrates that system performance is constrained primarily by electrical resistance and electromagnetic counter-effects rather than purely geometric or magnetic scaling.

Conclusions

The experimental results demonstrated a clear divergence from the initial theoretical predictions, indicating that the analytical model was insufficiently representative of the real operating conditions of the generator system. The primary source of this discrepancy arises from the omission or underestimation of key non-ideal effects, particularly Back EMF, resistive losses within the coil windings, and magnetic flux leakage across the air gap. These factors collectively reduced the effective induced voltage and introduced performance limitations that were not captured in the simplified theoretical framework.

In addition to electromagnetic losses, mechanical constraints played a significant role in limiting system efficiency. Rotor mass distribution and imbalance increased the torque required for operation and introduced vibration, which in turn affected the stability of the air gap and contributed to inconsistent electromagnetic coupling. Furthermore, the reliance on idealised rotational conditions did not account for structural deformation and alignment issues observed during testing, which further reduced overall performance.

The findings also highlight that coil resistance was a critical limiting parameter, restricting current flow and reducing usable output power despite increases in induced EMF at higher rotational speeds. Although increasing rotational velocity and magnetic flux density remains a valid route for improving performance, practical implementation is constrained by mechanical stability limits and material-related losses.

To improve future iterations of the design, the theoretical model should incorporate a more comprehensive treatment of Back EMF, dynamic load effects, and resistive heating to better predict real-world behaviour. Mechanically, improvements in rotor balancing, bearing stability, and structural rigidity would reduce vibration-induced losses and enhance energy transfer efficiency. Additionally, improved integration between electrical and mechanical subsystems through earlier-stage collaborative design would reduce incompatibilities observed during assembly. Overall, the study demonstrates that reliable generator performance cannot be achieved through electromagnetic optimisation alone, but requires a fully coupled electromechanical design approach supported by iterative experimental validation.

Reference

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Appendix A

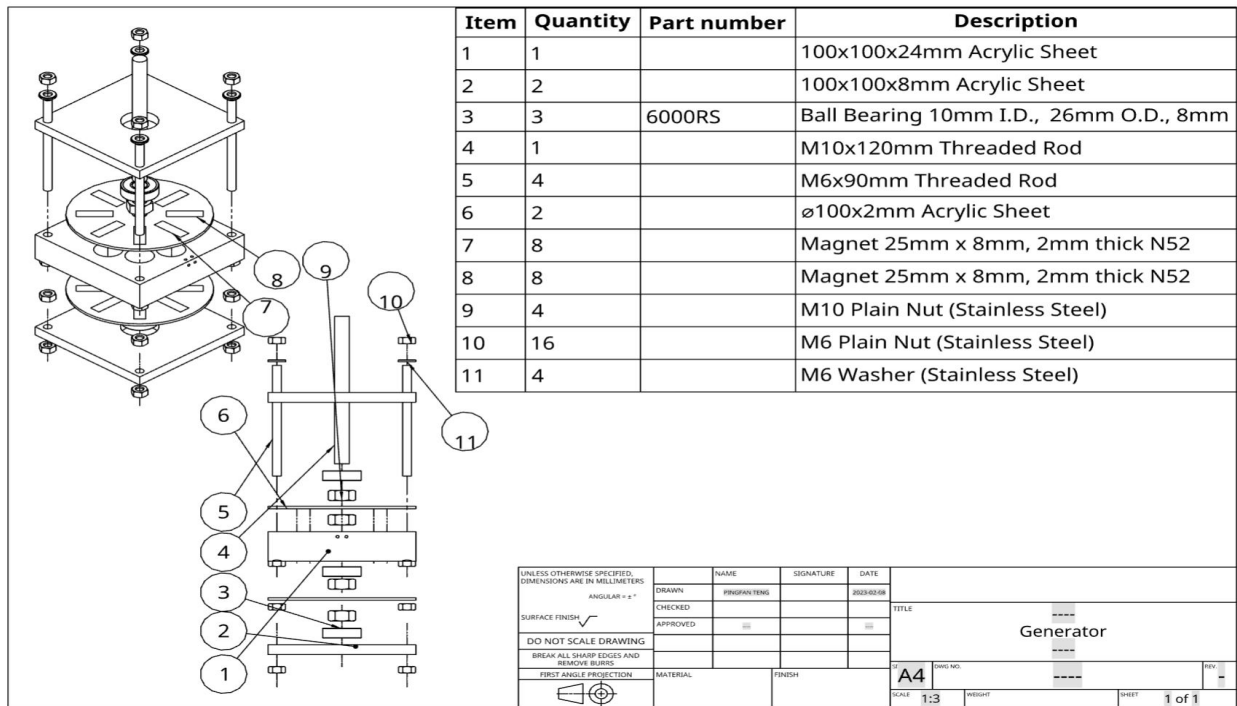


Figure A1: Bills of Material (BOM) of the AC Generator.



Figure A2: Voltage output of the AC Generator when tested at home with a handrill.